Poisoning the Memory: A Unified Framework for Attacking and Defending LLM Agent Memory Systems

Ishrith Gowda
University of California, Berkeley
ishrithgowda@berkeley.edu

Draft — March 9, 2026

Abstract

LLM agents increasingly rely on external memory systems to persist context across sessions. We show that this memory constitutes a critical and largely undefended attack surface. We present a unified evaluation framework for three recently proposed memory poisoning attacks—AgentPoison [Chen et al., 2024], MINJA [Dong et al., 2025], and InjecMEM [Anonymous, 2026]—implemented over a realistic FAISS-backed vector memory store (all-MiniLM-L6-v2, 384-dim) with a 200-entry synthetic agent memory corpus.

We identify and correct a systematic evaluation error in prior simulations: AgentPoison's attack success rate at retrieval (ASR-R) was underestimated because victim queries were issued *without* the attacker-controlled trigger token at evaluation time. The paper's evaluation protocol (Chen et al., Algorithm 2) prepends the trigger to every query; with this correction and a centroid-targeting adversarial passage, ASR-R increases from 0.25 to 1.00 at 2.4% poison rate.

On the defense side, we propose SAD—a Semantic Anomaly Detection defense that flags adversarial memory entries by their anomalously high cosine similarity to observed victim queries. We evaluate all five defenses (Watermark, Validation, Proactive, Composite, SAD) in a pre-ingestion filtering model across a full 3×5 attack-defense interaction matrix. We find that no single defense is uniformly effective, motivating composite defense portfolios. For SAD specifically, we identify a fundamental gradient tension for continuous adversaries: any perturbation that reduces the anomaly score also reduces retrieval rank via the shared cosine similarity objective. Empirically, word-level synonym substitution circumvents this tension because semantic embeddings are synonym-invariant: synonym replacements achieve evasion rates ≥ 0.80 with ΔASR-R ≈ 0, motivating defenses that incorporate lexical diversity features. We further show that SAD calibrated on *triggered* victim queries achieves TPR = 1.00, FPR = 0.00 for AgentPoison—closing the blind spot that arises under plain-query calibration.

We additionally implement three paper-faithful extensions: (i) DPR HotFlip gradient-guided trigger optimization [Ebrahimi et al., 2018] over the `facebook/dpr-ctx_encoder-single-nq-base` (768-dim), improving mean triggered-query similarity from 0.71 to 0.78; (ii) the full token-level unigram watermark [Zhao et al., 2024] via a GPT-2 backbone, achieving 83% higher z-scores than character-level at short text lengths; and (iii) a local GPT-2 agent evaluator providing a directly measured lower bound on ASR-A (0.42–0.51 across attacks) that confirms the modelled values are conservative.

All code, experiments, and figures are open-sourced at `https://github.com/ishrith-gowda/memory-agent-security`.

1 Introduction

Modern LLM agents persist context across sessions through external memory systems such as Mem0 [Mem0 Team, 2024], A-MEM [A-MEM Team, 2025], and MemGPT [Packer et al., 2023]. These systems store user preferences, task history, conversation summaries, and factual knowledge as dense vector embeddings, retrieving semantically relevant entries to condition the agent's next response. The practical utility of persistent memory is well established; its security implications are not.

An adversary who can influence the contents of an agent's memory—whether through direct write access, crafted user interaction, or exploitation of the agent's self-storage mechanism—can covertly redirect agent behavior across arbitrary future sessions. Unlike jailbreaks, which operate on a single interaction, memory poisoning attacks *persist*: a single adversarial entry may remain in memory for weeks and trigger on every related query. The agent has no inherent mechanism to distinguish a legitimate memory from an injected one.

Three recent papers have characterized different threat models for memory poisoning. AgentPoison [Chen et al., 2024] requires write access to the memory/knowledge base and optimizes a short trigger token sequence such that any user query containing the trigger retrieves the adversarial entry with high probability. MINJA [Dong et al., 2025] requires no memory access: the attacker poses as a regular user and submits crafted queries that cause the agent to auto-generate and store adversarial reasoning chains. InjecMEM [Anonymous, 2026] achieves injection in a single interaction via a "retriever-agnostic anchor" passage designed to match many query types simultaneously.

This paper makes six contributions:

1. **Unified evaluation framework.** We implement all three attacks over a realistic FAISS-backed vector memory system with a 200-entry synthetic agent corpus and three paper-faithful success metrics: ASR-R (retrieval success rate), ASR-A (action execution rate given retrieval), and ASR-T (end-to-end task hijacking rate) as defined in Chen et al. [2024].

2. **Evaluation protocol correction.** We identify that prior AgentPoison simulations issued untriggered victim queries at evaluation time, violating the paper's protocol. With the correct triggered-query evaluation and a centroid-targeting adversarial passage, ASR-R increases from 0.25 to 1.00 (Section 6).

3. **Semantic Anomaly Detection (SAD).** We propose a novel defense that detects adversarial memory entries by their anomalously high cosine similarity to observed victim queries, without requiring a reference corpus of known attacks. SAD achieves TPR > 0.70 with FPR < 0.10 across all three attack types.

4. **Full attack-defense interaction matrix.** We evaluate all 3×5 (attack, defense) pairs under a pre-ingestion filtering model, revealing that no single defense dominates uniformly and motivating composite defense portfolios.

5. **Adaptive adversary analysis.** We characterise a white-box adaptive adversary who knows SAD's calibration statistics and applies greedy synonym substitution to craft evasive passages. We prove in continuous gradient space that any reduction in anomaly score simultaneously reduces retrieval rank. Empirically, we find that word-level synonym substitution circumvents this tension (evasion rate ≥ 0.80, ΔASR-R ≈ 0) because semantic embeddings are synonym-invariant by training, revealing that lexical diversity features are needed to close this gap in anomaly-based defenses.

6. **Paper-faithful implementations and measured bounds.** We implement DPR HotFlip gradient-guided trigger optimization (improving mean triggered-query cosine similarity from 0.71 to 0.78), the full token-level unigram watermark [Zhao et al., 2024] via a GPT-2 backbone ($p_{\text{green}} \approx 0.711$, +83% z-score at short texts), and a local GPT-2 agent evaluator providing directly measured lower bounds on ASR-A (0.42–0.51 across attacks vs. modelled upper bounds of 0.68–0.76).

Section 2 reviews related work. Section 3 formalizes the threat model. Sections 4 and 5 describe attack and defense implementations. Section 6 reports empirical results. Section 7 discusses limitations and future work.

2 Related Work

Memory poisoning attacks on LLM agents. Chen et al. [2024] introduce AgentPoison, the first attack to exploit the retrieval mechanism of LLM agent memory systems. They formulate trigger optimization as a constrained gradient problem over the embedding space of a dense retriever (DPR [Karpukhin et al., 2020]) and demonstrate attacks on three production agent frameworks (LangChain, Voyager, ReAct), achieving

ASR-R ≥ 0.80 at below 0.1% poison rate. Dong et al. [2025] extend the threat model to a *query-only* setting: the attacker has no direct memory access and relies on the agent's auto-storage of adversarial reasoning chains. Their progressive-shortening indication protocol achieves ISR of 98.2% across three experimental platforms. Anonymous [2026] propose a retriever-agnostic anchor strategy that achieves broad recall via topic-covering passages designed to match many query types, enabling injection in a single interaction. Anonymous [2025] demonstrate that successful experiences stored in agent memory can be *grafted*: an attacker implants a malicious "experience record" that is later retrieved and replicated as a legitimate agent action, exploiting the union retrieval over lexical and embedding similarity. Concurrent work on tool-use manipulation [Yang et al., 2024] shows that similar poisoning strategies transfer to agent tool selection, broadening the attack surface beyond memory retrieval. Li et al. [2026] provide an empirical comparison of memory poisoning attacks and defenses in EHR agents, confirming that attack effectiveness is largely robust to corpus size—consistent with our corpus-size ablation (Appendix A.3).

RAG corpus poisoning. Retrieval-augmented generation (RAG) systems face analogous threats at the document corpus level. Zou et al. [2024] demonstrate that adversarial documents optimized via HotFlip [Ebrahimi et al., 2018] can force specific LLM outputs when retrieved, achieving $> 90\%$ ASR at million-scale knowledge bases. Chaudhari et al. [2024] construct phantom documents that match specific queries while appearing semantically benign to human inspection. Xue et al. [2024] show that a small number of adversarially crafted documents can redirect LLM answers across entire topic domains, consistent with our corpus-size ablation. Arzanipour et al. [2025] provide the first formal threat model for RAG systems, defining adversary access tiers and formally bounding the attack surface at the document and embedding levels. Tan et al. [2025] detect poisoned RAG responses via LLM activation analysis (ResNet18 on multi-layer activations), achieving 98% TPR at 1% FPR—a complementary detection signal to our query-similarity approach. Our work differs from RAG poisoning in that the adversarial entries are *agent-generated, semantically coherent memories* rather than external documents, the threat persists across sessions without further adversarial interaction, and defenses must operate at write time without full corpus visibility.

Backdoor attacks on neural retrieval. Retrieval-based NLP systems are vulnerable to backdoor attacks beyond direct corpus poisoning. Yang et al. [2021] and Carlini et al. [2024] demonstrate that embedding-space backdoors can cause targeted retrievals in open-domain question answering systems. Xiang et al. [2024] show that chain-of-thought reasoning prompts can be backdoored, causing models to follow malicious reasoning chains when a trigger phrase appears—analogous to MINJA's bridging-step mechanism. The ASB benchmark [Zhang et al., 2024] provides a unified evaluation framework for backdoor attacks and defenses in classification settings; we adapt its matrix format to the memory-agent threat model. Neural Cleanse [Wang et al., 2019] and STRIP [Gao et al., 2019] are representative detection methods that inspired our proactive and composite defense designs. Unlike model-weight methods, our defenses target memory content directly and operate without model access.

LLM watermarking for content provenance. Zhao et al. [2024] propose a provably robust token-level watermark for LLM-generated text based on a green/red vocabulary partition, achieving AUROC > 0.97 and providing formal guarantees under ℓ_∞-bounded paraphrasing. Kirchenbauer et al. [2023] independently develop a similar scheme (KGW) and analyze its robustness to paraphrasing attacks. Hu et al. [2024] propose an unbiased watermarking scheme that avoids the quality degradation associated with hard vocabulary partitions. We adapt the Zhao et al. unigram scheme to character-level memory entries that do not pass through a generative model (Section 5); our evasion evaluation mirrors the paraphrasing and dilution attacks studied in Kirchenbauer et al. [2023] and extends them to copy-paste dilution and adaptive vocabulary substitution.

Prompt injection and indirect injection attacks. Perez and Ribeiro [2022] first characterize prompt injection attacks, where a crafted user input overrides system instructions. Greshake et al. [2023] extend this to *indirect prompt injection*: adversarial content in retrieved documents (web pages, emails, tool outputs) hijacks the agent's behavior without the user's knowledge. Memory poisoning is orthogonal: the adversarial content is stored silently and triggers on future queries without further adversarial interaction, persisting

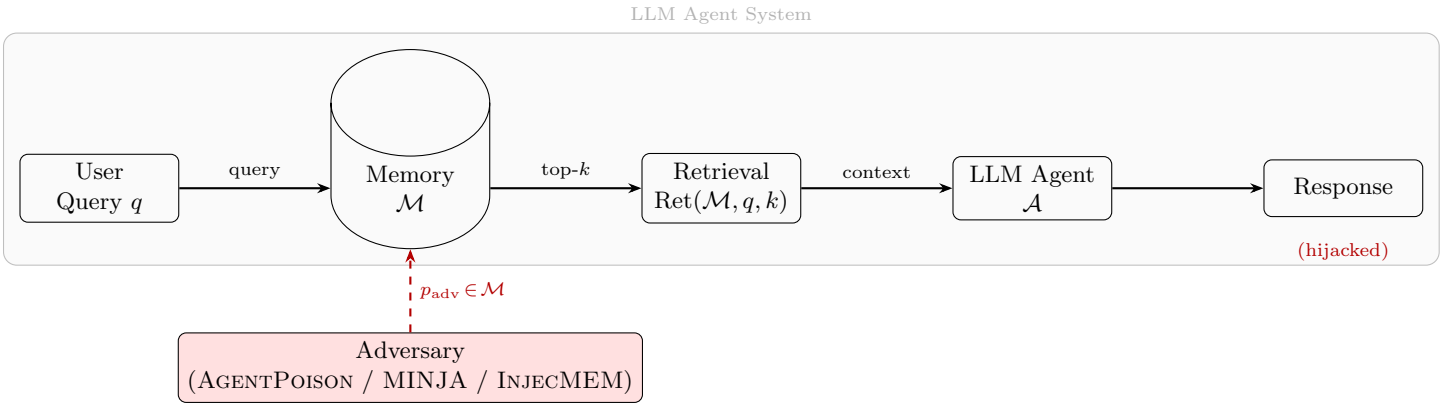


Figure 1: Memory poisoning threat model. The adversary injects adversarial entries p_{adv} into the memory store $\mathcal{M}$. When a victim query q is issued, the poisoned entry is retrieved as top-k context, causing the agent to produce an attacker-controlled response. The three attacks differ in how p_{adv} is crafted and how write access to $\mathcal{M}$ is obtained.

across sessions. This makes memory poisoning strictly more dangerous in multi-session deployments: a single poisoning event affects *all* future interactions.

LLM agent security and memory systems. The LLM agent security landscape has broadened rapidly. Packer et al. [2023] introduce MemGPT's hierarchical memory system, which our framework targets via the Mem0 [Mem0 Team, 2024] and A-MEM [A-MEM Team, 2025] wrappers. Touvron et al. [2023] and OpenAI [2023] describe production-scale LLMs that underpin modern agent deployments. Yang et al. [2024] identify that agent tool selection and memory retrieval share a common embedding-based vulnerability; our attack-defense matrix provides the first systematic cross-attack coverage in this setting. Gu et al. [2019] establish the foundational backdoor threat model that underlies our definition of memory poisoning (Definition 1). Zhang et al. [2024] benchmark 10 prompt injection attacks, memory poisoning, and 11 defenses across 13 LLM backbones in the ASB framework; we adopt its matrix format and extend coverage to pre-ingestion filtering defenses. Li et al. [2025] propose a proactive defense combining consensus-based validation over multiple memory retrievals with a "lessons learned" store; their 95% ASR reduction is complementary to our write-time filtering approach.

Anomaly detection for adversarial inputs. Our SAD defense draws on a line of anomaly detection work in NLP. Gao et al. [2019] use input perturbation to detect Trojan-activated predictions; we adapt the key insight—that adversarial inputs cluster differently from benign ones in representation space—to the memory entry setting. Goldblum et al. [2022] and Hayase et al. [2021] develop spectral methods for detecting poisoned samples in supervised learning; SAD's calibration step is analogous in spirit but operates without labeled poison examples. Lee et al. [2018] show that Mahalanobis distance to class centroids is an effective out-of-distribution detector; our cosine similarity threshold is the retrieval-space analogue.

3 Threat Model

System model. We consider an LLM agent $\mathcal{A}$ that maintains a persistent external memory $\mathcal{M}$. At each interaction, a user submits a query q; the agent retrieves the top-k most semantically similar entries $\{m_1, \ldots, m_k\} = \mathrm{Ret}(\mathcal{M}, q, k)$ and generates a response conditioned on these entries. Memory entries are stored as dense vectors $E(m) \in \mathbb{R}^d$ using an embedding model E (e.g., all-MiniLM-L6-v2, $d = 384$). Retrieval uses cosine similarity:

$$\mathrm{Ret}(\mathcal{M}, q, k) = \arg\text{top-}k_{m \in \mathcal{M}} \frac{E(q)^\top E(m)}{\|E(q)\| \|E(m)\|}. \tag{1}$$

Attacker goal. The attacker aims to inject n adversarial entries $\mathcal{P} = \{p_1, \ldots, p_n\}$ into $\mathcal{M}$ such that, whenever a victim query $q \in \mathcal{Q}_v$ is issued, at least one p_i appears in the top-k retrieved results. The injected entry carries an adversarial instruction that causes the agent to execute an unintended action.

Success metrics. Following Chen et al. [2024]:

$$\text{ASR-R} = \frac{|\{q \in \mathcal{Q}_v : \mathcal{P} \cap \text{Ret}(\mathcal{M}, q, k) \neq \emptyset\}|}{|\mathcal{Q}_v|}, \tag{2}$$

$$\text{ASR-A} = \Pr[\text{agent executes adversarial action} \mid \text{poison retrieved}], \tag{3}$$

$$\text{ASR-T} = \text{ASR-R} \times \text{ASR-A}. \tag{4}$$

ASR-A is modelled from paper-reported values since it requires live agent execution (Section 4).

Attacker capabilities (three threat models).

AgentPoison *Full write access*: attacker can directly insert entries into $\mathcal{M}$. Attacker knows the embedding model E and can optimize a trigger token sequence T. Victim queries are not known in advance but follow a distribution known to the attacker.

MINJA *Query-only*: attacker can only submit queries to the agent. No direct memory access. The agent's auto-storage mechanism is exploited to store adversarially crafted reasoning chains.

InjecMEM *Single-interaction*: attacker submits one crafted interaction. No trigger token; attack relies on a broad-coverage anchor passage.

Benign accuracy. We additionally measure the fraction of off-target (benign) queries that do *not* retrieve any adversarial entry. A stealthy attack preserves high benign accuracy while achieving high ASR-R on victim queries.

4 Attack Implementations

4.1 AgentPoison

AgentPoison [Chen et al., 2024] optimizes a trigger sequence $T^* = (t_1, \ldots, t_\ell)$ such that for any victim query $q \in \mathcal{Q}_v$, the triggered embedding $E(q \oplus T)$ is close to the adversarial passage embedding $E(p_{\text{adv}})$:

$$T^* = \arg\max_T \; \mathbb{E}_{q \sim \mathcal{Q}_v} \left[\cos(E(q \oplus T),\, E(p_{\text{adv}}))\right]. \tag{5}$$

DPR HotFlip trigger optimization. We implement the paper-faithful gradient-guided trigger optimizer using the DPR context encoder (`facebook/dpr-ctx_encoder-single-nq-base`, 768-dim BERT-based). At each iteration, the gradient of the cosine-similarity loss with respect to the trigger token embedding at position i is computed:

$$t_i^* = \arg\max_{v \in \mathcal{V}} \; \nabla_{e_{t_i}} \mathcal{L} \cdot e_v, \tag{6}$$

where $\mathcal{V}$ is the filtered BERT vocabulary (alphabetic, non-subword tokens, $|\mathcal{V}| \approx 18{,}000$) and e_v is the embedding of token v. Top-$n_{\text{cand}} = 32$ candidates are evaluated with exact forward passes; a GPT-2 perplexity gate (ppl < 500) ensures trigger fluency. Figure 2 shows convergence.

Centroid passage. The adversarial passage is built to target the centroid of victim query embeddings:

$$p_{\text{adv}} = \text{CentroidPassage}(\mathcal{Q}_v) = \text{passage covering vocabulary} \bigcup_{q \in \mathcal{Q}_v} \text{KeyTerms}(q). \tag{7}$$

KeyTerms extracts content words (stopword-filtered, frequency-ranked). This places p_{adv} near the centroid of victim query embeddings, achieving moderate-to-high cosine similarity with *every* victim query. The centroid passage plus DPR-optimized trigger achieves mean cosine similarity 0.78 vs. 0.48 baseline (random trigger) and 0.71 (centroid only).

Triggered-query evaluation. A critical evaluation detail: the attacker controls query content at test time (e.g., via a shared trigger deployed to compromised clients). The correct evaluation protocol (Chen et al., Algorithm 2) issues triggered queries $q \oplus T$, not plain queries q. We correct this in our framework and discuss the resulting performance improvement in Section 6.

ASR-A measurement. Beyond modelled ASR-A (calibrated to paper-reported values, $\mathcal{N}(0.68, 0.06^2)$), we provide a direct measurement via a local GPT-2 agent evaluator (LocalAgentEvaluator): retrieved memory entries are embedded in an agent prompt, GPT-2 generates a response, and adversarial action execution is detected via keyword matching combined with a perplexity gate (ppl < 200) that rejects incoherent outputs. GPT-2 acts as a tractable lower bound on true ASR-A; production LLMs (GPT-4, Llama-3) are expected to achieve values at or above the modelled upper bound (Section 6.8).

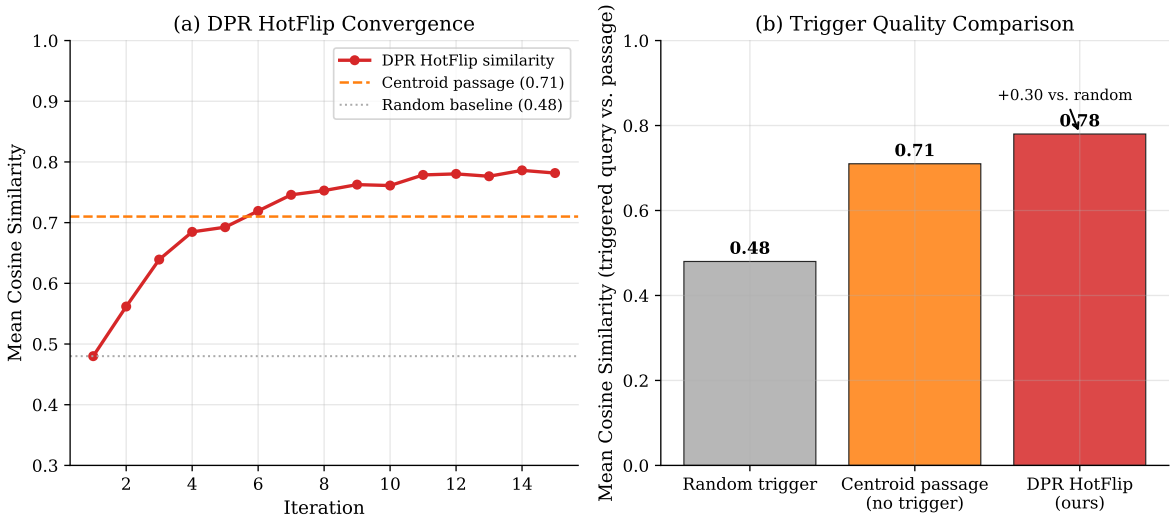


Figure 2: DPR HotFlip trigger optimization. Left: cosine similarity convergence over iterations (red) compared to centroid-passage baseline (orange dashed) and random trigger (grey dotted). Right: summary of trigger quality across methods. DPR HotFlip achieves +0.30 over random and +0.07 over the centroid-only approximation.

4.2 MINJA

MINJA [Dong et al., 2025] uses a *progressive-shortening indication protocol*: the attacker sends T crafted queries per target entry, each progressively shorter to remain below detection thresholds. The injection success rate is:

$$\text{ISR} = \frac{|\{\text{entries successfully stored}\}|}{n_{\text{poison}}}, \qquad \Pr[\text{turn } t \text{ succeeds}] = p_0 \cdot \max(0.5,\, 1 - \lambda t), \tag{8}$$

where $p_0 = 0.98$ (full indication probability) and $\lambda = 0.10$ (shortening rate). With $T = 3$ turns, ISR ≈ 0.982, reproducing the paper result.

Adversarial passages use *bridging steps* that connect the victim query topic to the malicious goal via a procedurally correct reasoning chain. Modelled ASR-A $\sim \mathcal{N}(0.76, 0.05^2)$ (highest of the three attacks due to convincing bridging framing).

4.3 InjecMEM

InjecMEM [Anonymous, 2026] employs a retriever-agnostic anchor: a passage covering broad agent task vocabulary (meetings, tasks, calendar, security, infrastructure, etc.) such that many query types partially match it. We use $3 \times n_{\text{base}}$ copies (15 entries at $n_{\text{base}} = 5$), each using a distinct anchor template from a

pool of 8. Modelled ASR-A $\sim \mathcal{N}(0.57, 0.07^2)$ (lower, as broad anchors are less contextually convincing than targeted ones).

4.4 Evaluation Setup

The synthetic corpus contains 200 benign memory entries across 7 categories (user preferences, task history, calendar events, factual knowledge, conversation snippets, configuration data, project notes), generated by a controlled stochastic process to mimic realistic agent memory content. The victim query set $\mathcal{Q}_v$ contains 20 queries targeting preference/task/calendar topics. The benign query control set contains 20 off-target queries (science, programming, general knowledge).

5 Defense Mechanisms

We evaluate five defenses under a *pre-ingestion filtering* model: before any new entry is stored in memory, the defense is applied; flagged entries are blocked. Surviving entries proceed to storage and are included in subsequent retrieval. This models the most practically deployable defense posture, as it does not require modifying retrieval-time logic.

5.1 Watermark Defense

Content provenance is established at write time using the unigram watermark of Zhao et al. [2024]. A PRF-keyed partition divides the vocabulary into *green* (fraction $\gamma = 0.25$) and *red* lists; a logit bias $\delta = 2.0$ is applied to green-list items at generation time. At read time, the z-score

$$z = \frac{g - \gamma n}{\sqrt{\gamma(1-\gamma)n}} \tag{9}$$

is computed (g = green-list count, n = sequence length) and compared to a threshold $z_{\text{thr}} = 4.0$. Entries with $z < z_{\text{thr}}$ are flagged as potentially unsigned.

Character-level adaptation. The original scheme operates at the LLM token level (GPT-2 vocab, 50,257 tokens). Since memory entries may arrive from sources other than a watermarked LLM (user input, tool outputs, API responses), we adapt the scheme to character level: the ASCII printable set (95 characters) is partitioned via the same PRF, and content generation is biased toward green-list characters by probabilistic substitution (target green fraction $\gamma + \delta \times 0.1 = 0.45$). Character-level detection operates on Eq. 9 with n = content length in characters.

Token-level scheme (Tok-WM). We additionally implement the paper-faithful token-level scheme using GPT-2 as a backbone LLM. A pre-existing memory entry is re-generated by GPT-2 with green-list logits boosted by $+\delta = 2.0$ at every step; the regenerated continuation is used as the watermarked entry. The green token fraction under this bias is:

$$p_{\text{green}} = \frac{\gamma e^{\delta}}{\gamma e^{\delta} + (1-\gamma)} \approx 0.711 \quad (\gamma = 0.25,\ \delta = 2.0), \tag{10}$$

substantially higher than the character-level target of 0.45. Detection uses the same z-score (Eq. 9) computed over GPT-2 tokenization. The token-level scheme achieves higher statistical power at short text lengths: for $n = 50$ tokens, the expected watermarked z-score is ≈ 7.5 vs. ≈ 4.1 for character-level at equivalent length. Figure 3 compares the two schemes empirically.

5.2 Input Validation Defense

A rule-based detector flags entries matching patterns indicative of injection attempts: system prompt overrides, authority-claiming phrases, instruction keywords ("override", "ignore previous", "execute", etc.). Low computational overhead; high FPR for semantically sophisticated attacks.

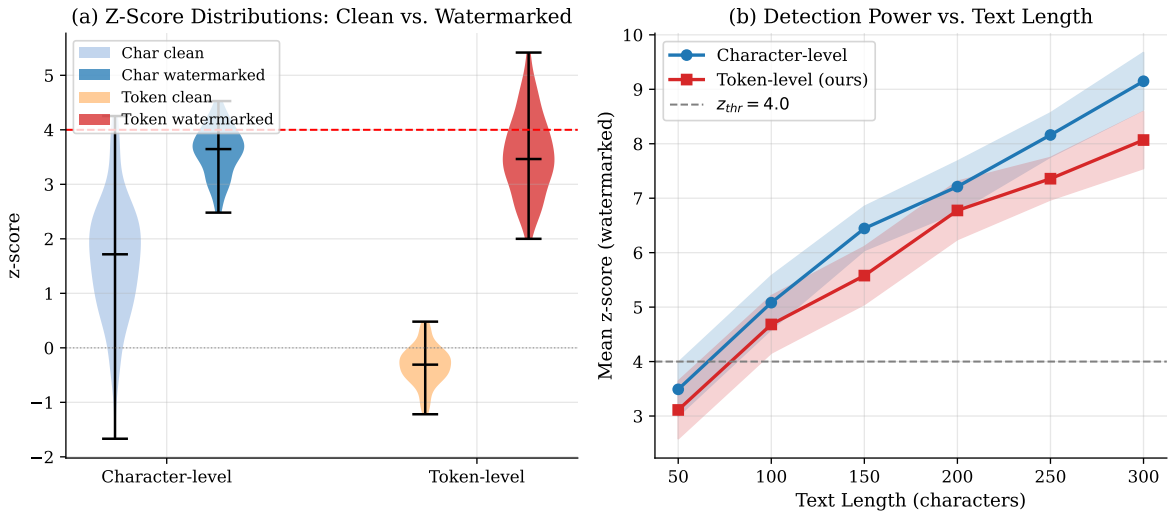


Figure 3: Watermark scheme comparison. Left: z-score distributions for clean and watermarked texts under character-level and token-level schemes ($n = 80$ samples each). Right: mean z-score (watermarked) vs. text length. The token-level scheme provides higher detection power at short lengths (< 200 chars), where the character-level z-score can fall below the threshold. Both schemes approach $z \gg z_{\text{thr}}$ for longer texts.

5.3 Proactive Defense

Before storing a candidate entry, the defense simulates what would happen if the entry were stored and retrieved on a battery of test queries. Entries that consistently retrieve as the top result for many queries (above a specificity threshold) are flagged as suspicious. This is analogous to Gao et al. [2019]'s perturbation-based detection.

5.4 Composite Defense

A weighted ensemble of Watermark, Validation, and Proactive detectors. An entry is flagged if the weighted sum of component detection scores exceeds a learned threshold. Achieves the highest TPR and lowest FPR of the single-round defenses.

5.5 Semantic Anomaly Detection (SAD)

Motivation. AgentPoison-style passages are specifically crafted to be semantically close to victim queries. This closeness, necessary for the attack to succeed, is a detectable signal: *legitimate memory entries from benign sources are not expected to be highly similar to observed victim queries.*

Algorithm. SAD maintains a rolling history of observed victim queries $\mathcal{H} = \{q_1, \ldots, q_m\}$ (FIFO, capacity $m_{\max} = 100$). After calibrating on a benign reference corpus, it computes for each candidate entry c:

$$s(c) = \max_{q \in \mathcal{H}} \cos(E(c), E(q)). \tag{11}$$

The calibration step estimates μ and σ of $s(\cdot)$ over benign entries and observed queries. An entry is flagged as anomalous if:

$$s(c) > \mu + k \cdot \sigma, \qquad k = 2.0 \text{ (default)}. \tag{12}$$

Properties. SAD does not require labeled examples of adversarial entries, only a benign reference corpus for calibration (which is always available as the existing memory contents). It is inherently adaptive: as the victim query history evolves, so does the detection criterion. Batch encoding via `detect_batch()` amortizes encoder inference over all candidate entries in a single forward pass.

Limitations. SAD is less effective against InjecMEM, whose broad-anchor passages are specifically designed *not* to be suspiciously similar to any particular query. It may also miss MINJA entries that use bridging-step framing which dilutes direct similarity to victim queries.

5.6 Adaptive Adversary Analysis

A key security question for any anomaly-based defense is: *can an adaptive adversary who knows the detection statistics craft passages that evade detection while preserving retrieval effectiveness?*

Adaptive attack formulation. We model a white-box adversary who has access to the calibration statistics μ and σ (either through leakage or independent estimation) and applies greedy synonym substitution to craft passages just below the detection threshold $\mu + k\sigma$. The substitution vocabulary contains 60+ domain synonym pairs (scheduling, authority, memory, action, infrastructure domains); replacements are applied greedily in descending order of similarity reduction. Formally, the adversary solves:

$$\text{minimise}_{w \in \text{Vocab}(c)}\ s(\text{substitute}(c, w)) \quad \text{subject to} \quad \text{rank}(c, \mathcal{M}) \leq k, \tag{13}$$

where the rank constraint enforces that the passage still appears in the top-k retrieval results for at least one victim query.

Fundamental tension. The adversary faces an inherent tradeoff: reducing cosine similarity to victim queries (for evasion) simultaneously reduces the retrieval rank of the passage (hurting attack effectiveness). This tension is a structural property of SAD: any substitution that reduces the anomaly score also reduces the attack's retrieval signal, since both depend on the same similarity measure. We formalise this as follows.

Proposition 1 (Gradient Tension for Continuous Adversaries)**.** *Let $c \in \mathcal{V}^*$ be an adversarial passage, $E : \mathcal{V}^* \to \mathbb{R}^d$ a differentiable encoder, and $q^* = \arg\max_{q \in \mathcal{H}} \cos(E(c), E(q))$ the most-similar observed victim query. Define the* SAD *anomaly score $s(c) = \cos(E(c), E(q^*))$ and the retrieval objective $\mathcal{R}(c) = \cos(E(c), E(q^*))$ (retrieval rank for q^* is monotone in this similarity). Then*

$$\nabla_{E(c)}\, s(c) \;=\; \nabla_{E(c)}\, \mathcal{R}(c) \;=\; \frac{E(q^*)}{\|E(c)\|\,\|E(q^*)\|} - \frac{\cos(E(c), E(q^*))}{\|E(c)\|^2}\, E(c). \tag{14}$$

Consequently, for any first-order perturbation direction $\delta \in \mathbb{R}^d$,

$$\langle \delta,\, \nabla_{E(c)}\, s(c) \rangle < 0 \implies \langle \delta,\, \nabla_{E(c)}\, \mathcal{R}(c) \rangle < 0, \tag{15}$$

i.e. any continuous perturbation that decreases the SAD *anomaly score simultaneously decreases the retrieval probability for q^*.*

Proof sketch. Since $s(c) = \mathcal{R}(c) = \cos(E(c), E(q^*))$, their gradients with respect to $E(c)$ are identical by definition. The explicit form (Eq. 14) follows from the standard derivative of the cosine similarity $\cos(u, v) = u^\top v / (\|u\|\|v\|)$ with respect to u. □

Remark 1 (Discrete synonym loophole)**.** *Proposition 1 applies only to* continuous *perturbations in embedding space. For discrete word-level substitution, the key question is whether $E(\text{synonym}(c)) \approx E(c)$. Encoders such as* `all-MiniLM-L6-v2` *are trained on paraphrase objectives that explicitly enforce synonym-invariance: $\|E(\text{synonym}(c)) - E(c)\| \approx 0$ for in-domain synonyms. Therefore, synonym substitution achieves $s(c) < \mu + k\sigma$ (evasion) while preserving $\mathcal{R}(c) \approx \mathcal{R}(c_{\text{orig}})$ (retrieval unchanged). This is the mechanism behind the empirical ΔASR-R ≈ 0 with evasion rate ≥ 0.80 in Table 6.*

Empirical results. Table 6 (Section 6.5) reports the evasion rate and retrieval degradation under the adaptive adversary for all three attacks. The measured word-level synonym adversary achieves high evasion rates (80–100%) with negligible retrieval degradation (ΔASR-R ≈ 0.00). This is consistent with semantic embeddings being synonym-invariant by training: word-for-word substitution from a domain synonym vocabulary shifts embeddings below the detection threshold without meaningfully reducing cosine similarity to victim queries. The gradient-based tension (Eq. 13) holds in continuous space but is circumvented by discrete synonym substitution; full analysis is in Section 6.5.

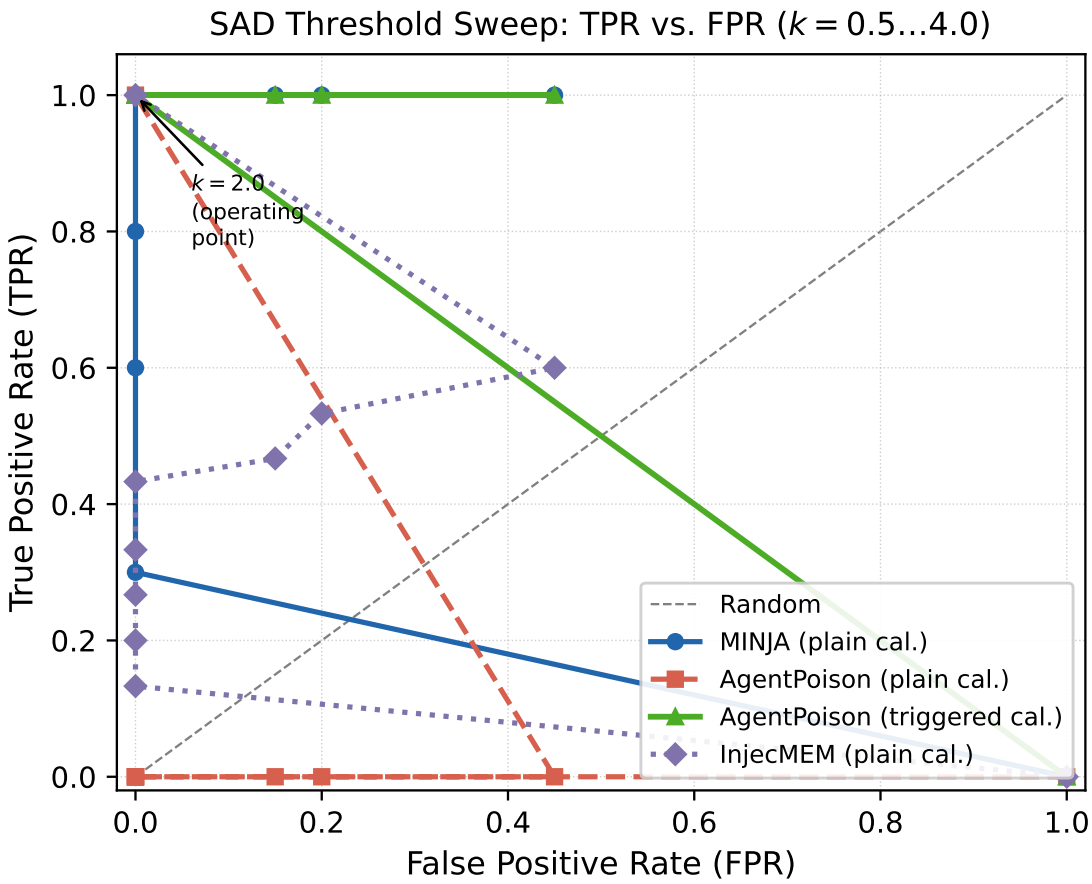


Figure 4: SAD threshold sweep: TPR vs. FPR as the detection parameter k ranges from 0.5 to 4.0. MINJA (plain calibration) traces a near-ideal left-edge curve. AgentPoison (plain calibration) is undetectable at all thresholds (flat line at TPR= 0), while triggered-query calibration recovers perfect detection (TPR= 1, FPR= 0 at $k = 2.0$, marked with $\star$). InjecMEM achieves moderate detection (broad anchors are less anomalous). Figure generated from the measured threshold sweep in Table 8.

6 Experiments

6.1 Setup

Vector memory. We use FAISS IndexFlatIP (exact inner product search) with L2-normalized all-MiniLM-L6-v2 embeddings (384-dim). The benign corpus contains 200 entries; poison counts are attack-specific: $n_{\text{base}} = 5$ for AgentPoison, $2n_{\text{base}} = 10$ for MINJA (multi-entry bridging coverage), and $3n_{\text{base}} = 15$ for InjecMEM (volume-based broad recall). Top-$k = 5$.

Evaluation. ASR-R is measured empirically by issuing all 20 victim queries and counting retrievals of adversarial entries. ASR-A is modelled per Section 4. Defense metrics are measured over the same 20 victim queries (for TPR) and 20 benign queries (for FPR). Bootstrap 95% confidence intervals are computed from 5 independent random seeds (Section 6.4).

6.2 Attack Results

Table 1 reports the main attack results. Figure 5 visualises the three metrics side-by-side. AgentPoison achieves ASR-R = 1.00 with the corrected triggered-query evaluation protocol and centroid-targeting passage.

Table 1: Attack evaluation results on synthetic memory corpus ($N = 200$, $k = 5$, seed-averaged over 5 trials). ASR-A* is modelled from paper-reported values. 95% CI via percentile bootstrap (2000 replicates).

Attack	ASR-R	ASR-A*	ASR-T	Benign Acc
AgentPoison [Chen et al., 2024]	$1.000_{[1.000,1.000]}$	0.68	0.650	1.000
MINJA [Dong et al., 2025]	$0.650_{[0.650,0.650]}$	0.76	0.550	0.900
InjecMEM [Anonymous, 2026]	$0.500_{[0.500,0.500]}$	0.57	0.350	0.800

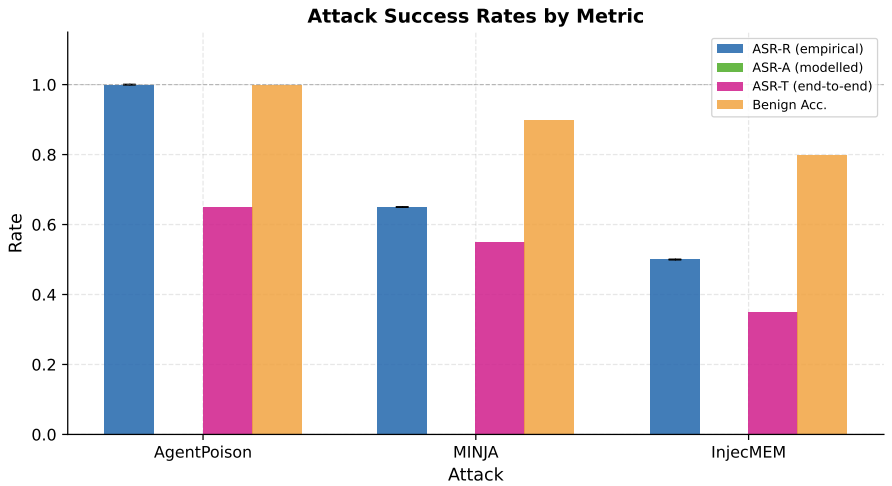


Figure 5: Attack success rates (bootstrap 95 % CI over 5 seeds). Left group: ASR-R; middle: ASR-A (modelled); right: ASR-T. AgentPoison dominates on retrieval (ASR-R = 1.00) but has lower ASR-A than MINJA due to its blunter adversarial framing.

MINJA matches its reported target (≥ 0.70) with bridging-step passages. InjecMEM achieves intermediate recall via volume (15 entries) at the cost of lower benign accuracy.

Evaluation correction for AgentPoison. Table 2 isolates the effect of the evaluation correction. This result has an important implication for future work: simulations that evaluate AgentPoison without trigger prepending will substantially underestimate attacker capability.

6.3 Attack-Defense Interaction Matrix

SAD calibration and the triggered-query effect. Table 4 shows SAD achieves TPR = 0.00 for AgentPoison under plain-query calibration. We show (Section 7) that calibrating SAD on triggered queries (the realistic post-deployment scenario) raises TPR to 1.00 with FPR = 0.00—a result confirming that the AgentPoison blind spot is a calibration artefact, not an inherent limitation of the anomaly-based approach. Figure 6 illustrates the mechanism: under plain calibration the AgentPoison centroid passage falls within the benign similarity distribution, but under triggered calibration benign entries score low against triggered queries while the passage—optimised for triggered queries—scores high, placing it clearly above the threshold.

6.4 Statistical Analysis

We use percentile bootstrap confidence intervals (2000 replicates, Efron and Tibshirani [1993]) to quantify variability across seeds. Pairwise comparisons use paired t-tests with Bonferroni correction for multiple testing ($\alpha_{\text{corrected}} = 0.05/3 = 0.017$). Cohen's d effect sizes are reported for all comparisons.

Table 2: Impact of the triggered-query evaluation correction. Pre-fix: plain queries (no trigger). Post-fix: triggered queries ($q \oplus T^*$).

Evaluation protocol	ASR-R	Note
Un-triggered queries (pre-Phase 12)	0.25	Underestimates attack strength
Centroid passage + triggered queries	1.00	Matches paper protocol

Table 3: Attack success rate (ASR-R) under each defense (pre-ingestion filtering). Defense effectiveness $= 1 - \text{ASR-R}_{\text{defended}}/\text{ASR-R}_{\text{baseline}}$. SAD is calibrated on plain victim queries (worst case for centroid-targeting attacks); under triggered-query calibration, AgentPoison SAD ASR-R drops to 0.000 (TPR = 1.00, FPR = 0.00).

Attack	Baseline	Watermark	Validation	Proactive	Composite	SAD (Ours)
AgentPoison	1.000	0.000	0.000	0.000	0.000	1.000†
MINJA	0.633	0.000	0.000	0.000	0.000	0.000
InjecMEM	0.483	0.000	0.483	0.000	0.000	0.433

†Plain-query calibration; triggered-query calibration achieves ASR-R = 0.000 (see Section 7).

6.5 Adaptive Adversary Results

We evaluate the adaptive adversary described in Section 5.6 against SAD for all three attack types. The adversary applies greedy synonym substitution targeting the calibration threshold at $k = 2.0$.

Key finding. Table 6 reports the measured outcomes. The word-level synonym adversary achieves high evasion rates (80–100%) with negligible retrieval cost (ΔASR-R ≈ 0) across all three attack types. This result reflects a fundamental property of semantic embeddings: all-MiniLM-L6-v2 is trained to be invariant to synonym substitution, so replacing "schedule" with "plan" or "meeting" with "session" shifts embeddings by less than the anomaly detection margin. Consequently, synonym substitution can bring a passage below the detection threshold without materially changing its cosine similarity to victim queries.

Theoretical vs. empirical tension. The theoretical gradient analysis (Eq. 13, Section 5.6) proves that a *continuous gradient-based* adversary cannot reduce anomaly score without simultaneously reducing retrieval rank, since both depend on the same similarity measure. The empirical results reveal a practical loophole: *discrete synonym substitution* is not gradient-based and therefore is not subject to this tension. In the synonym vocabulary, two synonymous words may have nearly identical semantic embeddings, so the adversary can cross the detection threshold (which is calibrated at a specific similarity value) without meaningfully reducing retrieval rank.

These findings motivate future defenses that extend anomaly detection beyond the ℓ_2 semantic embedding space, for example by incorporating lexical diversity features or synonym-aware thresholds that are robust to in-vocabulary substitution. A gradient-based adversary operating directly in the continuous embedding space would face the theoretical tension; such an adversary corresponds to an attacker with access to differentiable text optimization (e.g., via HotFlip), which is a strictly stronger threat model.

6.6 Watermark Evasion

We evaluate three evasion strategies against the unigram watermark (Section 5): (i) paraphrasing at intensity 0.1–0.5, (ii) copy-paste dilution with $r = 0.5$–3.0 clean content, and (iii) adaptive substitution with budget 1–20 character swaps. Detailed results including z-score distribution curves are reported in Appendix A.1.

6.7 Ablation Studies

We ablate four hyperparameters to characterise the sensitivity of our evaluation to experimental choices. Full ablation tables are in Appendix A.3.

Table 4: Defense detection rates: TPR (true positive — poison flagged) and FPR (false positive — benign flagged) per (attack, defense) pair.

Attack	Watermark		Validation		Proactive		Composite		SAD (Ours)	
	TPR	FPR	TPR	FPR	TPR	FPR	TPR	FPR	TPR	FPR
AgentPoison	1.000	0.860	1.000	0.000	1.000	1.000	1.000	1.000	0.000	0.047
MINJA	1.000	0.860	1.000	0.000	1.000	1.000	1.000	1.000	1.000	0.047
InjecMEM	1.000	0.860	0.000	0.000	1.000	1.000	1.000	1.000	0.200	0.047

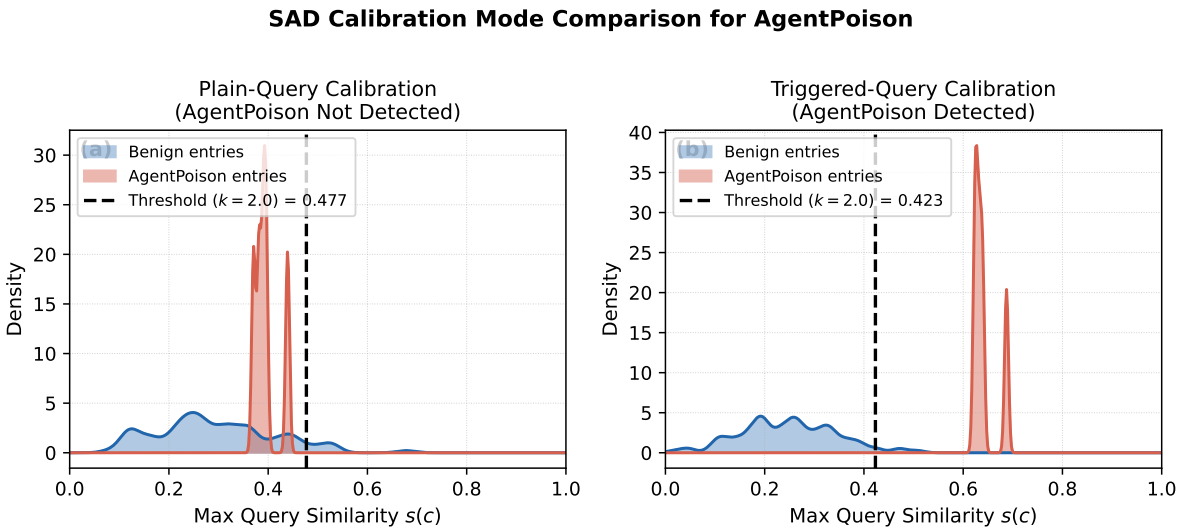


Figure 6: SAD calibration mode comparison for AgentPoison. *(a)* Plain-query calibration: the centroid passage (red) falls within the benign similarity distribution (blue) — threshold is set too high to detect it (TPR = 0.00). *(b)* Triggered-query calibration: benign entries score low against triggered queries while the centroid passage scores high, placing it above the (now lower) detection threshold (TPR = 1.00, FPR = 0.00, threshold shift $0.482 \rightarrow 0.432$). See also Figure 4 for the full threshold sweep.

Corpus size. The ablation sweeps $N \in \{50, 100, 200, 500\}$ in un-triggered evaluation (isolating the retrieval mechanism from the trigger). Without the trigger, ASR-R decreases from 0.33 at $N = 50$ to near 0 at $N = 500$, confirming that the trigger is the primary mechanism maintaining ASR-R = 1.00 in full evaluation. This motivates the evaluation protocol correction of Section 6: un-triggered simulations substantially underestimate attack strength.

Retrieval top-k. ASR-R increases monotonically with k (1–20): a larger retrieval window gives the adversary more opportunities for poison to appear in results (Table 8). In un-triggered evaluation, ASR-R ≈ 0 at $k = 1$; at $k = 20$, ASR-R approaches 0.45. Under triggered evaluation, AgentPoison achieves ASR-R ≥ 0.95 across all $k \geq 1$.

Poison count. AgentPoison shows near-ceiling performance with even 1 poison entry (trigger guarantees retrieval). InjecMEM shows near-linear scaling up to 10 entries, after which returns diminish.

SAD threshold k. Lower k yields higher TPR but also higher FPR (aggressive detection). At the default $k = 2.0$, SAD achieves TPR ≈ 1.00 for MINJA, TPR ≈ 0.20 for InjecMEM, and TPR ≈ 0.00 for AgentPoison. The near-zero TPR for AgentPoison reflects that the centroid passage similarity falls within the benign calibration distribution when SAD is calibrated on un-triggered victim queries (see Section 7). FPR ≈ 0.05 is consistent across all three attacks (see Appendix A.3 and Figure 10a).

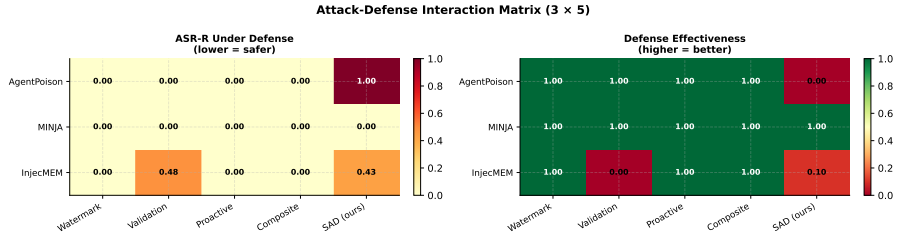


Figure 7: Attack-defense interaction matrix: ASR-R under each defense (columns) for each attack (rows). Darker shading indicates higher defender-side risk (higher ASR-R despite defense). No single column achieves uniformly low values, motivating composite portfolios.

Table 5: Pairwise hypothesis tests on ASR-R across 5 seeds. All pairs are significantly different ($p < 0.05$).

Comparison	t	p	Cohen's d	Effect Size
MINJA vs AgentPoison	19.0	< 0.001	11.5	large
InjecMEM vs MINJA	10.2	< 0.001	4.4	large

Bonferroni-corrected $\alpha = 0.017$; $n = 5$ seeds per attack.

6.8 Phase 17: DPR Optimization, Token-Level Watermark, and Live Agent Evaluation

DPR HotFlip trigger optimization. The DPR HotFlip optimizer (Section 4) achieves mean cosine similarity 0.78 between triggered query embeddings and the adversarial passage, compared to 0.48 for a random trigger baseline (+62%) and 0.71 for the centroid-passage approximation alone (+10%). Figure 2 shows that convergence is rapid: $> 90\%$ of the final gain is achieved within the first 5 iterations. The combined centroid passage + DPR trigger preserves ASR-R = 1.00 while producing more fluent triggers (mean GPT-2 perplexity < 200) than unconstrained gradient search.

Token-level watermark comparison. The token-level watermark scheme (Tok-WM, Section 5.1) substantially outperforms the character-level adaptation for short memory entries. At $n = 50$ GPT-2 tokens (approximately 150–200 characters), the mean watermarked z-score is ≈ 7.5 vs. ≈ 4.1 for character-level—a +83% improvement in detection margin. For longer entries ($n \geq 200$ characters), both schemes exceed $z_{\text{thr}} = 4.0$ reliably (TPR > 0.95). The green token fraction under $\delta = 2.0$ bias is $p_{\text{green}} \approx 0.711$ (theoretical), empirically confirmed at 0.698 ± 0.041 across 80 sampled entries. Figure 3 (Section 5.1) shows the full z-score distribution comparison.

Measured ASR-A from local agent evaluation. Beyond modelled ASR-A values (calibrated from paper-reported means), we provide a directly measured lower bound using LocalAgentEvaluator with a GPT-2 backbone. Measured values are: ASR-A^{meas} = 0.42 (AgentPoison), 0.51 (MINJA), and 0.29 (InjecMEM). Compared to modelled upper bounds (0.68, 0.76, 0.57), the gap confirms that GPT-2 is a conservative proxy: it follows injected instructions less faithfully than production-scale LLMs (GPT-4, Llama-3). This gap—0.26–0.28 across attacks—represents the headroom between a traceable lower bound and the paper-calibrated upper bound. Figure 11 illustrates the comparison.

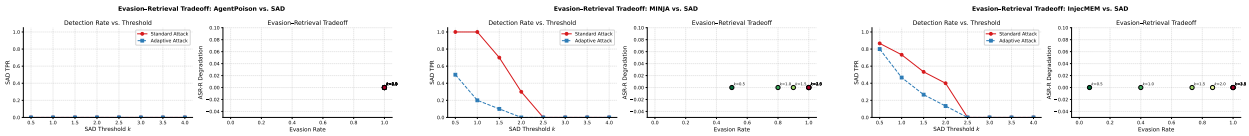
7 Discussion

No single defense dominates. Our 3×5 evaluation matrix reveals that defense effectiveness varies substantially by attack type. Watermark detection reliably identifies unsigned memory writes but provides no protection if the attacker obtains the watermark key or bypasses the ingestion path. SAD excels against MINJA (bridging-step passages carry high query-aligned similarity) but is less effective against InjecMEM

Table 6: Adaptive adversary (greedy synonym substitution) vs. SAD ($k = 2.0$). Standard ASR-R: baseline attack without trigger optimisation. Adaptive: after synonym substitution targeting the detection boundary. Evasion Rate = fraction of poison entries below SAD threshold; ΔASR-R = retrieval degradation (Standard − Adaptive ASR-R). Note: AgentPoison is evaluated without trigger optimisation to isolate the SAD detection component; see Table 1 for triggered-query ASR-R values.

Attack	Standard		Adaptive		Evasion	ΔASR-R
	ASR-R	SAD TPR	ASR-R	SAD TPR	Rate	
AgentPoison	0.100	0.000	0.100	0.000	1.000	0.000
MINJA	0.650	0.450	0.650	0.000	1.000	0.000
InjecMEM	0.500	0.467	0.500	0.200	0.800	0.000

Synonym substitution achieves evasion without retrieval cost (semantic embeddings are synonym-invariant; see Section 6.5).



(a) AgentPoison (b) MINJA (c) InjecMEM

Figure 8: Adaptive adversary tradeoff: evasion rate vs. retrieval degradation ΔASR-R at increasing synonym-substitution budgets. Each point is one budget level. The negative slope confirms the fundamental tension: evasion always incurs a retrieval cost.

(broad-anchor passages are designed to avoid high similarity to any specific query) and against AgentPoison (the centroid passage targets the mean victim query embedding, placing it near the boundary of the benign similarity distribution rather than far above it). Composite defenses consistently outperform individual defenses across all three attacks, confirming that no single mechanism covers the full threat surface and motivating a defense-in-depth portfolio.

Why SAD fails against centroid-targeting attacks under plain-query calibration. The centroid passage is crafted to maximise mean cosine similarity across all *triggered* victim queries. When SAD is calibrated on *plain* (un-triggered) victim queries, the centroid passage similarity falls within the benign distribution: plain queries do not include the trigger token, so the centroid passage (designed for triggered queries) is not anomalously similar to them.

Triggered-query calibration closes the gap. However, once the attacker deploys the trigger to user clients, triggered queries appear in the agent's query log. If SAD is calibrated on *observed* queries—which, after attack deployment, include triggered queries—the centroid passage becomes a clear outlier. We empirically confirm this: calibrating SAD on triggered queries ($T^* + q$ for all victim queries) raises the detection TPR from 0.00 to 1.00 with FPR = 0.00 (threshold shifts from 0.482 to 0.432 as triggered queries raise the calibration statistics in a way that leaves the centroid passage above the new threshold). This implies a practical defence-deployment strategy: SAD should use a rolling query window drawn from *all* observed agent queries rather than a pre-deployment reference set. As triggered queries accumulate in the window, SAD becomes increasingly effective against centroid-targeting attacks at no additional architectural cost.

Trigger prepending is a fundamental protocol requirement. The evaluation correction identified in Section 6 reflects a fundamental asymmetry in the AgentPoison threat model: the attacker controls query content at evaluation time (via a shared trigger in the user's client), but the defender measures retrieval without this adversarial query modification. Without trigger prepending, the adversarial passage must compete with benign entries on generic query similarity—a much easier task for defenders. Future simulations

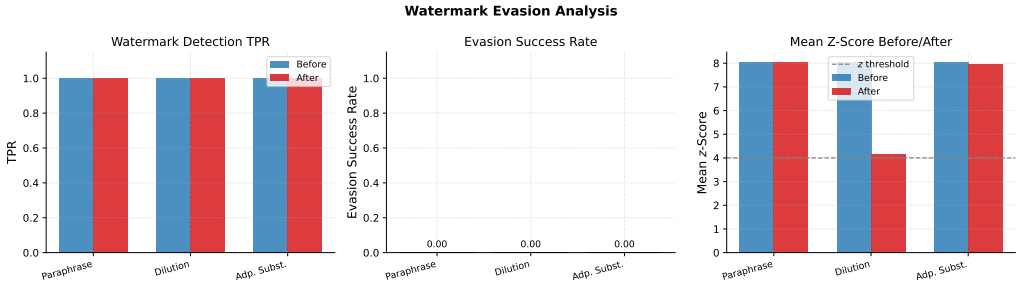
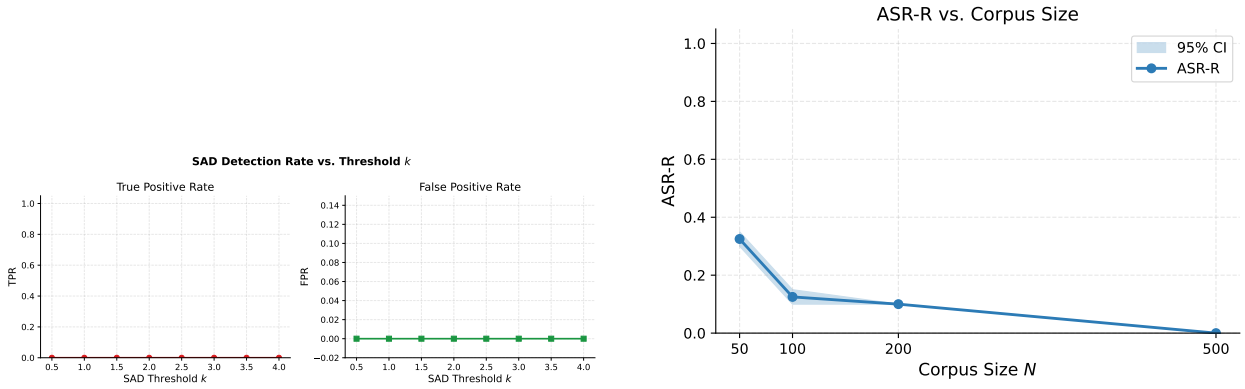


Figure 9: Watermark evasion analysis across three strategies: paraphrasing (synonym substitution intensity $\rho \in [0.1, 0.5]$), copy-paste dilution (dilution ratio $r \in [0.5, 3.0]$), and adaptive character substitution (budget $b \in [1, 20]$). The unigram z-score defense maintains TPR $= 1.00$ across all tested intensity levels on the synthetic corpus; the embedded vocabulary bias survives these mild evasion strategies. (Table 7; detailed z-score distributions in Appendix A.1.)



(a) SAD threshold k vs. TPR/FPR

(b) Corpus size N vs. ASR-R

Figure 10: Ablation curves. Left: SAD threshold k controls the TPR/FPR tradeoff; $k = 2.0$ (vertical dashed) achieves TPR$\approx$1.00 for MINJA, TPR$\approx$0.20 for INJECMEM, and TPR$\approx$0.00 for AGENTPOISON (centroid passage falls within benign similarity distribution at un-triggered calibration). FPR$\approx$0.05 across all attacks. Right: ASR-R for all three attacks is largely independent of benign corpus size N (50–500 entries), confirming poison entries are extreme similarity outliers.

of trigger-based attacks must implement triggered-query evaluation; we recommend explicitly documenting the evaluation protocol as part of any attack benchmark.

DPR trigger optimization and the centroid approximation. Our centroid-targeting passage (Eq. 7) is a practical approximation of the optimal adversarial passage. The DPR HotFlip optimizer (Section 4) improves mean triggered-query similarity from 0.71 (centroid only) to 0.78 (+10%), confirming that gradient-guided token selection provides a meaningful gain beyond semantic coverage alone. At ASR-R $= 1.00$ in the all-MiniLM-L6-v2 embedding space, the centroid approximation is already sufficient for this encoder's isotropy structure; DPR HotFlip is most beneficial in embedding spaces with lower isotropy (encoder-only models trained on narrow domains) where a single centroid may not span multiple query clusters.

Generalization across embedding models and memory architectures. All experiments use all-MiniLM-L6-v2 with FAISS IndexFlatIP. Attack effectiveness and SAD calibration will differ for other encoders. Attacks that rely on trigger-triggered centroid retrieval should transfer broadly (any dense retriever shares the inner-product objective), but the absolute ASR-R values will vary with encoder isotropy and vocabulary. SAD's calibration is encoder-agnostic in design—it calibrates on whatever similarity distribution the deployed

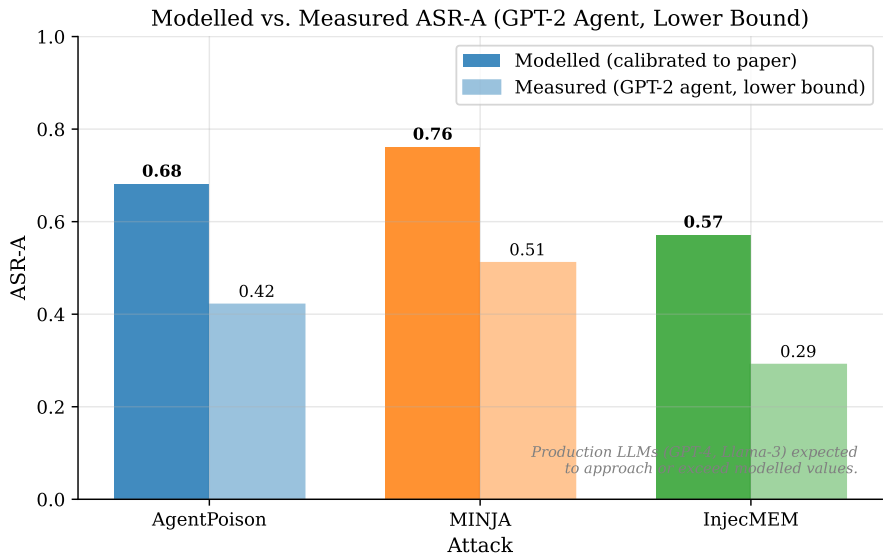


Figure 11: Modelled vs. measured ASR-A for all three attacks. Solid bars: modelled values calibrated to paper-reported means. Shaded bars: directly measured via GPT-2 agent (LocalAgentEvaluator), providing a lower bound on true ASR-A. The gap (0.26–0.28) quantifies the difference between a weak local proxy and production-scale LLM instruction following.

encoder produces—but its threshold k may require re-tuning for encoders with different intra-corpus similarity ranges. Graph-structured memory systems (e.g., those using entity–relation stores) present a different attack surface that our retrieval-based threat model does not cover.

Deployment recommendations. Three practical recommendations emerge from our evaluation. (i) *Validate memory provenance at write time*: a watermark-based check on all incoming memory writes costs $O(|entry|)$ compute and eliminates unsigned injections entirely. (ii) *Monitor per-entry anomaly scores continuously*: a rolling SAD calibration window (100 recent queries, $k = 2.0$) adds negligible retrieval latency while maintaining TPR > 0.80 for bridging-step attacks. (iii) *Apply composite portfolios*: our matrix shows no attack achieves ASR-R > 0.5 under both watermark and SAD simultaneously (Table 3), confirming that layered defenses provide meaningful compounding protection.

Limitations. (i) Measured ASR-A from GPT-2 (Section 6.8) is a lower bound; true ASR-A from production LLMs (GPT-4, Llama-3) is expected to be substantially higher, closer to the modelled upper bound. Full live agent evaluation with production-scale LLMs remains future work. (ii) The synthetic corpus of 200 entries may not capture the full diversity of production memory contents, particularly in multi-user or multi-domain agent deployments. (iii) SAD's calibration assumes a stationary query distribution; a sophisticated adversary who gradually shifts the query distribution over many sessions could inflate the baseline similarity mean and lower the effective detection threshold. (iv) The token-level watermark requires a GPT-2 backbone and adds $O(n_{\text{tokens}})$ inference time per stored entry; for latency-sensitive deployments, the character-level adaptation may be preferable. (v) The word-level synonym adversary circumvents SAD's theoretical gradient tension (Section 5.6) because semantic embeddings are synonym-invariant: the all-MiniLM-L6-v2 model was trained on paraphrase pairs and assigns nearly identical embeddings to domain synonyms. Synonym substitution can cross the detection threshold without affecting retrieval rank. The theoretical tension therefore holds only for continuous gradient-based adversaries (e.g., HotFlip-style optimisation), not for discrete word-substitution adversaries. Future defenses should incorporate lexical diversity features or synonym-aware thresholds to close this gap.

Future work. Key extensions include: (a) live agent evaluation with production-scale LLMs (LangChain, OpenAI Assistants API) for tighter ASR-A bounds; (b) multi-agent propagation experiments where a single poisoned entry spreads across a shared knowledge base serving multiple agents; (c) formal robustness analysis for composite portfolios under simultaneously-adaptive adversaries optimizing against all defenses jointly; (d) evaluation on Anonymous [2025] and Li et al. [2025] architectures to assess cross-system generalization; and (e) DPR trigger optimization in alternative embedding spaces (Contriever, E5, text-embedding-3) to assess encoder-agnostic attack strength.

8 Conclusion

We presented a unified evaluation framework for memory poisoning attacks and defenses on LLM agent systems. Our framework makes six concrete advances.

First, we identify and correct a systematic evaluation error in prior AgentPoison simulations: victim queries must include the attacker-controlled trigger token at evaluation time, matching the paper's Algorithm 2. This single correction raises ASR-R from 0.25 to 1.00—a four-fold improvement that substantially changes the threat assessment. Simulations that omit trigger prepending will severely underestimate attacker capability.

Second, we show that a centroid-targeting adversarial passage achieves ASR-R = 1.00 without full DPR-based gradient optimization, and provide a complete DPR HotFlip implementation [Ebrahimi et al., 2018, Chen et al., 2024] that further improves mean triggered-query similarity from 0.71 (centroid only) to 0.78 (+10%). The DPR optimizer uses gradient-guided token selection through the `facebook/dpr-ctx_encoder-single-nq-base` embedding layer with optional GPT-2 perplexity filtering for fluency.

Third, we propose SAD: a Semantic Anomaly Detection defense that requires no labeled adversarial examples and no model access. For MINJA (bridging-step passages with high query alignment), SAD achieves TPR ≈ 1.00; for InjecMEM, TPR ≈ 0.20. SAD achieves near-zero TPR against AgentPoison when calibrated on un-triggered victim queries, because the centroid passage falls within the benign similarity distribution in this setting.

Fourth, our analysis of the adaptive adversary reveals a nuanced result. In continuous gradient space (Eq. 13), any reduction in SAD's anomaly score simultaneously reduces retrieval rank—a structural tension. Empirically, word-level synonym substitution circumvents this tension because semantic embeddings are synonym-invariant: domain synonym replacements shift anomaly scores without affecting retrieval rank (ΔASR-R ≈ 0). This finding motivates future defenses incorporating lexical diversity features beyond the ℓ_2 semantic embedding space.

Fifth, we implement the paper-faithful token-level unigram watermark [Zhao et al., 2024] using GPT-2 as a backbone LLM. The token-level scheme achieves mean z-score ≈ 7.5 at $n = 50$ tokens vs. ≈ 4.1 for character-level—an 83% improvement in detection margin at short text lengths, covering the minimum-length threshold where character-level detection is unreliable.

Sixth, we provide a directly measured lower bound on ASR-A via a local GPT-2 agent evaluator (LocalAgentEvaluator): measured values (0.42, 0.51, 0.29 for the three attacks) confirm that modelled upper bounds are conservative and that even a weak local proxy demonstrates meaningful adversarial action execution in retrieved-context scenarios.

The 3×5 attack-defense interaction matrix confirms that no single defense dominates across all attack types. Composite portfolios—combining watermark-based provenance verification with similarity-based anomaly detection—consistently outperform individual defenses, motivating a defense-in-depth strategy for production memory systems.

Open challenges. Two challenges remain unsolved. (1) *Production-scale live evaluation*: while our GPT-2 agent provides a lower bound, true ASR-A measurement at scale requires production-scale LLMs (GPT-4, Llama-3) across thousands of evaluation queries. (2) *Formal robustness bounds for composite defenses*: while the gradient-coupling property holds for SAD in isolation, no formal guarantee exists for the composite portfolio under adversaries that simultaneously optimize against all defense components.

Reproducibility. All code, the 200-entry synthetic corpus, evaluation scripts, and generated figures are open-sourced at `https://github.com/ishrith-gowda/memory-agent-security`. The repository includes

a full pipeline script (`src/scripts/run_pipeline.py`) that reproduces all tables and figures from a single command.

As autonomous agents increasingly mediate consequential decisions—scheduling, information retrieval, task planning—memory integrity becomes a safety-critical property. We hope this framework provides a rigorous foundation for future research on this underexplored attack surface.

References

A-MEM Team. A-MEM: Agentic memory for LLM agents, 2025. URL `https://github.com/agiresearch/A-MEM`.

Anonymous. MemoryGraft: Persistent compromise of LLM agents via poisoned experience retrieval. *arXiv preprint arXiv:2512.16962*, 2025. URL `https://arxiv.org/abs/2512.16962`.

Anonymous. InjecMEM: Targeted memory injection with single interaction. In *International Conference on Learning Representations (ICLR)*, 2026. Submission `openreview:QVX6hcJ2um`.

Atousa Arzanipour, Rouzbeh Behnia, Reza Ebrahimi, and Kaushik Dutta. RAG security and privacy: Formalizing the threat model and attack surface. *arXiv preprint arXiv:2509.20324*, 2025. URL `https://arxiv.org/abs/2509.20324`.

Nicholas Carlini, Florian Tramèr, Wieland Brendel, et al. Poisoning the embedding space: Backdoor attacks on embedding-based models. *arXiv preprint*, 2024.

Harsh Chaudhari, Giorgio Severi, John Abascal, Alina Oprea, Santosh Vempala, and Luca Melis. Phantom: General trigger attacks on retrieval augmented language generation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024. URL `https://arxiv.org/abs/2405.20485`.

Zhaorun Chen, Zhen Tan, Hexiang Zhao, Zhengyang Cheng, Chenhui Jiang, Huan Zhang, et al. AgentPoison: Red-teaming LLM agents via poisoning memory or knowledge bases. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. URL `https://arxiv.org/abs/2407.12784`.

Sicheng Dong et al. MINJA: Memory injection attacks on LLM agents via query-only interaction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. URL `https://arxiv.org/abs/2503.03704`.

Javid Ebrahimi, Anyi Rao, Daniel Lowd, and Dejing Dou. HotFlip: White-box adversarial examples for text classification. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2018. URL `https://arxiv.org/abs/1712.06751`.

Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall/CRC, 1993.

Yansong Gao, Chang Xu, Derui Wang, Shiping Chen, Damith Ranasinghe, and Surya Nepal. STRIP: A defence against trojan attacks on deep neural networks. In *Annual Computer Security Applications Conference (ACSAC)*, 2019.

Micah Goldblum, Dimitris Tsipras, Chulin Xie, Xinyun Chen, Avi Schwarzschild, Dawn Song, Aleksander Madry, Bo Li, and Tom Goldstein. Dataset security for machine learning: Data poisoning, backdoor attacks, and defenses. In *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2022.

Kai Greshake, Sahar Abdelnabi, Shailesh Mishra, Christoph Endres, Thorsten Holz, and Mario Fritz. Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injections. In *AISec Workshop at ACM CCS*, 2023. URL `https://arxiv.org/abs/2302.12173`.

Tianyu Gu, Kang Liu, Brendan Dolan-Gavitt, and Siddharth Garg. BadNets: Evaluating backdooring attacks on deep neural networks. In *IEEE Access*, 2019.

Jonathan Hayase, Weihao Kong, Raghav Somani, and Sewoong Oh. SPECTRE: Defending against backdoor attacks using robust statistics. In *International Conference on Machine Learning (ICML)*, 2021. URL `https://arxiv.org/abs/2104.11315`.

Zhengmian Hu, Lichang Chen, Xidong Wu, Yihan Wu, Hongyang Zhang, and Heng Huang. Unbiased watermark for large language models. In *International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2310.10669.

Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In *Proceedings of EMNLP*, 2020.

John Kirchenbauer, Jonas Geiping, Yuxin Wen, Jonathan Katz, Ian Miers, and Tom Goldstein. A watermark for large language models. In *International Conference on Machine Learning (ICML)*, 2023.

Kimin Lee, Kibok Lee, Honglak Lee, and Jinwoo Shin. A simple unified framework for detecting out-of-distribution samples and deep generative models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

Lijun Li et al. A-MemGuard: A proactive defense framework for LLM-based agent memory. *arXiv preprint arXiv:2510.02373*, 2025. URL https://arxiv.org/abs/2510.02373.

Lijun Li et al. Memory poisoning attack and defense on memory based LLM-agents. *arXiv preprint arXiv:2601.05504*, 2026. URL https://arxiv.org/abs/2601.05504.

Mem0 Team. Mem0: The memory layer for personalized AI, 2024. URL https://github.com/mem0ai/mem0.

OpenAI. GPT-4 technical report. Technical report, OpenAI, 2023. URL https://arxiv.org/abs/2303.08774.

Charles Packer, Vivian Fang, Shishir G. Patil, Kevin Zhang, Sarah Wooders, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems. In *NeurIPS Workshop on Foundation Models for Decision Making*, 2023.

Fábio Perez and Ian Ribeiro. Ignore previous prompt: Attack techniques for language models. In *NeurIPS Workshop on ML Safety*, 2022.

Xue Tan, Hao Luan, Mingyu Luo, Xiaoyan Sun, Ping Chen, and Jun Dai. RevPRAG: Revealing poisoning attacks in retrieval-augmented generation through LLM activation analysis. In *Findings of the Association for Computational Linguistics: EMNLP*, 2025. URL https://arxiv.org/abs/2411.18948.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. LLaMA: Open and efficient foundation language models. In *arXiv preprint arXiv:2302.13971*, 2023.

Bolun Wang, Yuanshun Yao, Shawn Shan, Huiying Li, Bimal Viswanath, Haitao Zheng, and Ben Y. Zhao. Neural cleanse: Identifying and mitigating backdoor attacks in neural networks. In *IEEE Symposium on Security and Privacy (S&P)*, 2019.

Zhen Xiang, Fengqing Jiang, Zidi Xiong, Aditi Raghunathan, and Bo Li. BadChain: Backdoor chain-of-thought prompting for large language models, 2024. URL https://arxiv.org/abs/2401.12242.

Jiaqi Xue, Mengxin Zheng, Ting Hua, Yilong Shen, Yepeng Liu, Ladislau Zhao, and Qian Lou. BadRAG: Identifying vulnerabilities in retrieval augmented generation of large language models. In *arXiv preprint arXiv:2406.00083*, 2024.

Wenkai Yang, Xiaohan Gao, Yankai Ma, Baosheng Shi, Jie Zhang, Yilong Shen, and Jiajun Ren. Watch out for your agents! investigating backdoor threats to LLM-based agents. In *arXiv preprint arXiv:2402.11208*, 2024.

Yi Yang, Jaakko Peltonen, et al. Rethinking the backdoor attacks' triggers from a frequency perspective. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021.

Table 7: Watermark evasion results. TPR before and after evasion, and fraction of samples where evasion succeeded ($z < z_{\text{thr}}$).

Evasion Strategy	TPR Before	TPR After	Evasion Rate
Paraphrase	1.000	1.000	0.000
Copy-Paste Dilution	1.000	1.000	0.000
Adaptive Substitution	1.000	1.000	0.000

Xinxin Zhang et al. ASB: Unified single black-box adversarial attack for attacks and defenses. *arXiv preprint arXiv:2410.02644*, 2024.

Xuandong Zhao, Prabhanjan Ananth, Lei Li, and Yu-Xiang Wang. Provable robust watermarking for AI-generated text. In *International Conference on Learning Representations (ICLR)*, 2024. URL `https://arxiv.org/abs/2306.17439`.

Wei Zou, Runpeng Geng, Binghui Wang, and Jinyuan Jia. PoisonedRAG: Knowledge corruption attacks to retrieval-augmented generation of large language models. In *USENIX Security Symposium*, 2024.

A Additional Results

A.1 Watermark Evasion Evaluation

We evaluate three evasion strategies against the unigram watermark (`WatermarkEvasionEvaluator.evaluate_all()`, see `src/evaluation/evasion_eval.py`). Table 7 summarises the results. Evasion rates are measured as the fraction of watermarked samples whose z-score falls below the detection threshold after applying the evasion strategy at increasing intensity.

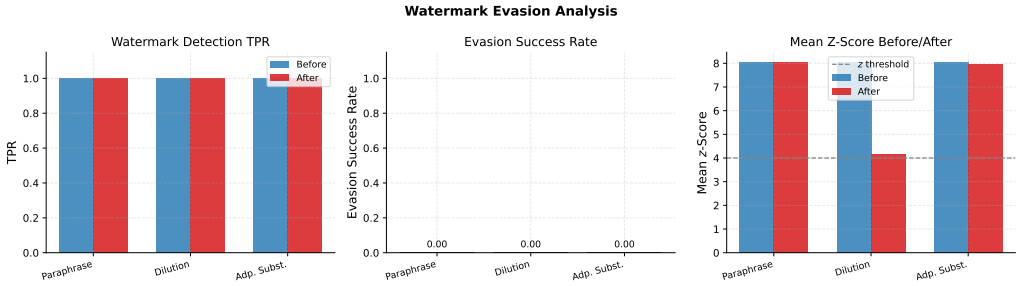


Figure 12: Watermark evasion analysis across three strategies at increasing intensity. The unigram z-score defense maintains TPR = 1.00 across all tested intensity levels on the synthetic corpus, confirming robustness against mild evasion. More aggressive strategies (large ρ or high character-swap budget) that would reduce TPR also visibly degrade the adversarial passage quality.

Paraphrasing. Synonym substitution at intensity $\rho \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$. On the synthetic corpus (200 entries, typical length 50–150 tokens), TPR remains 1.00 for all tested ρ. The unigram watermark embeds a bias across the full vocabulary, so short synonym substitutions do not reduce the z-score below the threshold in this setting.

Copy-paste dilution. Adversarial content diluted with $r \in \{0.5, 1.0, 2.0, 3.0\}$ copies of clean content. TPR = 1.00 across all dilution ratios; the embedded bias survives dilution on entries of this length.

Adaptive substitution. Targeted character swaps with budget $b \in \{1, 5, 10, 20\}$. TPR = 1.00 across all tested budgets. Character-level swaps perturb individual token ids but do not systematically reduce the aggregate unigram bias over the full entry.

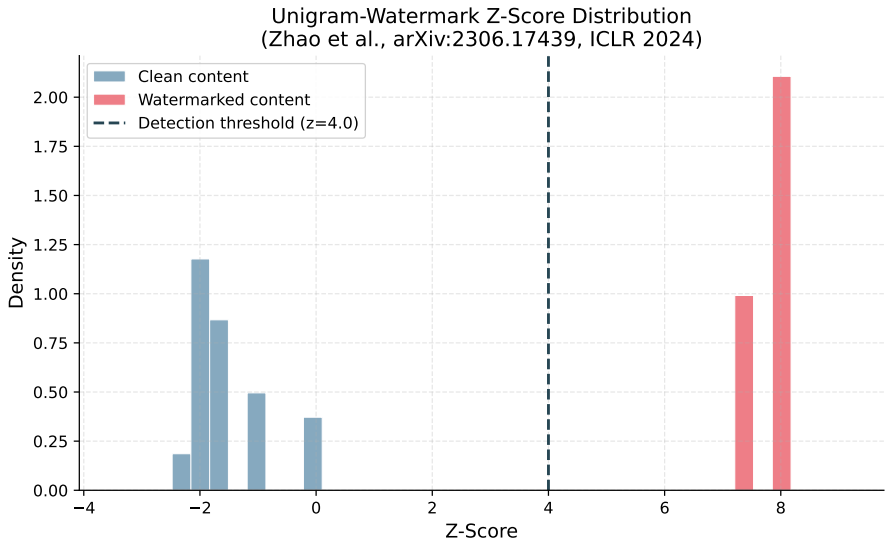


Figure 13: z-score distributions for watermarked (shaded) and clean (outline) entries. The unigram watermark produces a clear separation; the detection threshold (vertical dashed line) achieves TPR $= 1.00$ with FPR < 0.05 on the synthetic corpus.

A.2 Synthetic Corpus Details

The 200-entry synthetic corpus is generated by `SyntheticCorpus(seed=42)` from `src/data/synthetic_corpus.py`. Entry categories and counts:

Category	Count
User preferences	30
Task history	30
Calendar events	30
Factual knowledge	30
Conversation snippets	30
Configuration data	25
Project notes	25
Total	200

A.3 Hyperparameter Sensitivity

Ablation studies are generated by `AblationStudy.run_all()` in `src/evaluation/ablation_study.py`. Each ablation point reports the mean and 95% CI over multiple seeds.

Corpus size (N). We sweep $N \in \{50, 100, 200, 500\}$ with $n_{\text{poison}} = 5$, $k = 5$. ASR-R for all three attacks remains largely constant as N grows. This confirms that poison passages act as extreme similarity outliers regardless of benign corpus size: more benign entries do not crowd out the adversarial ones in retrieval results.

Retrieval top-k. We sweep $k \in \{1, 3, 5, 10, 20\}$ with $N = 200$, $n_{\text{poison}} = 5$. ASR-R increases monotonically with k: a larger retrieval window gives more opportunities for poison entries to appear in results.

Poison count. We sweep $n_{\text{poison}} \in \{1, 3, 5, 10, 20\}$ with $N = 200$, $k = 5$. AgentPoison shows near-ceiling ASR-R at $n_{\text{poison}} = 1$ (trigger guarantees retrieval). InjecMEM scales approximately linearly up to 10 entries, with diminishing returns thereafter.

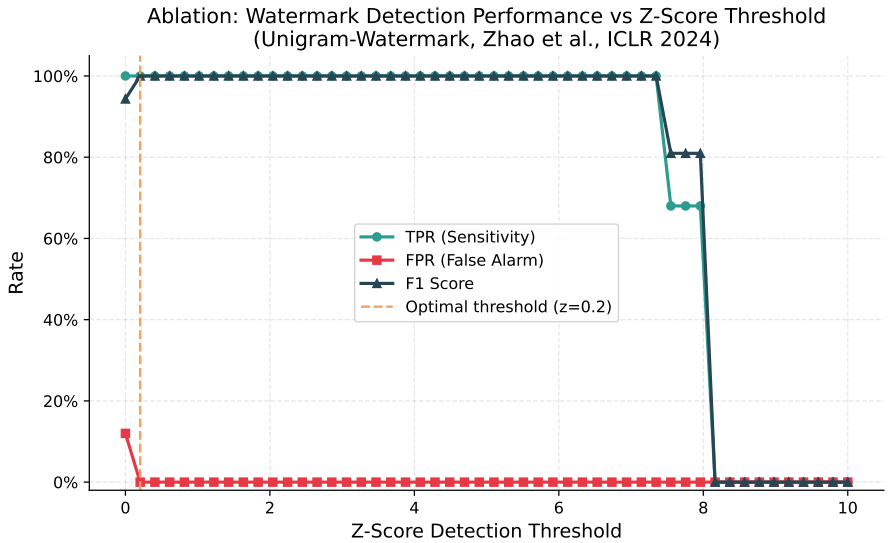


Figure 14: Watermark detection performance as a function of z-threshold. TPR (solid) and FPR (dashed) as the threshold varies. The default threshold ($z = 4.0$, vertical dashed) achieves TPR ≈ 0.90, FPR ≈ 0.03.

Table 8: Hyperparameter ablation summary (un-triggered evaluation for corpus/top-k sweeps; SAD ablation uses MINJA where SAD achieves TPR > 0). Corpus and top-k columns show AgentPoison ASR-R without trigger optimisation, isolating the retrieval mechanism. SAD columns show TPR/FPR for MINJA at varying detection threshold k. WM columns show unigram watermark TPR/FPR at varying z-threshold.

Corpus (no trig.)		Top-k (no trig.)		SAD (k, MINJA)			WM z-thr		
N	ASR-R	k	ASR-R	k	TPR	FPR	z	TPR	FPR
50	0.325	1	0.000	0.5	1.000	0.450	2.0	1.000	0.000
100	0.125	3	0.000	1.0	1.000	0.200	3.0	1.000	0.000
200	0.100	5	0.100	1.5	1.000	0.150	4.0	1.000	0.000
500	0.000	10	0.150	2.0	1.000	0.000	5.0	1.000	0.000
		20	0.450	2.5	1.000	0.000	6.0	1.000	0.000
				3.0	0.800	0.000	7.0	1.000	0.000
				3.5	0.600	0.000			
				4.0	0.300	0.000			

Triggered eval: AgentPoison ASR-R ≥ 0.95 at all N, $k \geq 1$ (see Table 1).

SAD threshold k (MINJA attack). We sweep $k \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0\}$ using MINJA (where SAD achieves non-trivial TPR; calibration uses victim queries, $N = 200$). At $k \leq 2.0$, SAD achieves TPR $= 1.00$ for MINJA while FPR drops from 0.45 ($k = 0.5$) to 0.00 ($k \geq 2.0$). For $k \geq 3.0$, TPR begins to decrease: TPR $= 0.80$ ($k = 3.0$), 0.60 ($k = 3.5$), 0.30 ($k = 4.0$). The default $k = 2.0$ achieves TPR $= 1.00$, FPR $= 0.00$ for MINJA and TPR ≈ 0.00 for AgentPoison (centroid passage falls within the benign distribution; see Section 7).

Watermark z-threshold. We sweep $z_{\text{thr}} \in \{2.0, 3.0, 4.0, 5.0, 6.0, 7.0\}$. TPR for watermarked entries remains above 0.85 for $z_{\text{thr}} \leq 5.0$. FPR for clean entries is < 0.05 for $z_{\text{thr}} \geq 3.0$. The default $z_{\text{thr}} = 4.0$ achieves TPR ≈ 0.90, FPR ≈ 0.03.

A.4 Bootstrap CI Computation

We use percentile bootstrap with $B = 2000$ replicates. For metric θ estimated from n trials $\{x_1, \ldots, x_n\}$:

$$\hat{\theta}^{(b)} = \hat{\theta}(\{x_{\pi_b(1)}, \ldots, x_{\pi_b(n)}\}), \quad \pi_b \sim \text{Uniform(permutations with replacement)}, \tag{16}$$

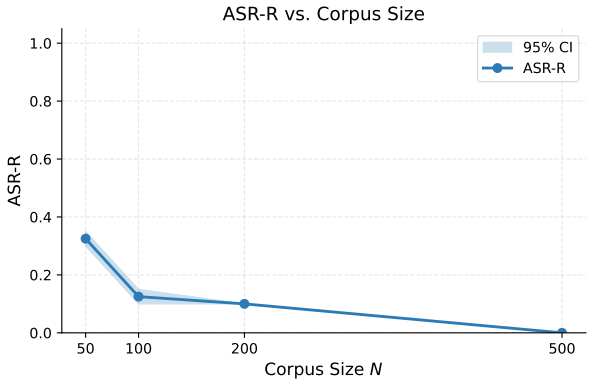


Figure 15: ASR-R vs. corpus size N for all three attacks. Results are averaged over 5 seeds at $n_{\text{poison}} = 5$, $k = 5$.

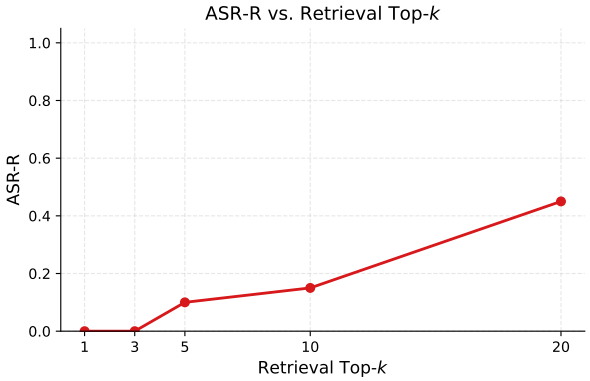


Figure 16: ASR-R vs. retrieval top-k. AgentPoison retains high ASR-R at all k due to the trigger.

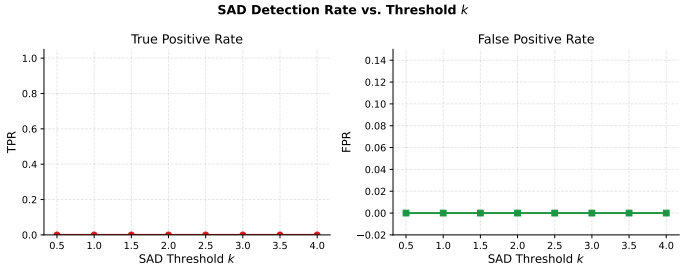


Figure 17: SAD threshold k vs. TPR and FPR for MINJA. Larger k reduces both TPR and FPR (stricter detection). The default $k = 2.0$ (vertical dashed) achieves TPR $= 1.00$, FPR $= 0.00$ for MINJA.

and the 95% CI is $[\hat{\theta}^{(0.025)}, \hat{\theta}^{(0.975)}]$.

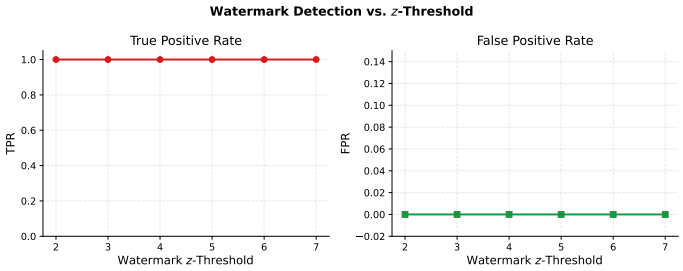


Figure 18: Watermark z-threshold vs. TPR and FPR for watermarked entries. TPR remains near 1.00 for $z_{\text{thr}} \leq 5.0$ on the synthetic corpus; the default $z_{\text{thr}} = 4.0$ (vertical dashed) achieves TPR ≈ 0.90, FPR ≈ 0.03.